

UST'-USA KOMI, Russian Federation

WMO: 234120

Lat: 65.9842N

Lon: 56.9058E

Elev: 79

StdP: 100.38

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.6	(c) -34.9	(d) -40.4	(e) 0.1	(f) -37.6	(g) -37.6	(h) 0.1	(i) -34.8	(j) 8.7	(k) -11.0	(l) 7.8	(m) -11.9	(n) 2.0	(o) 20	(p) 0.756

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	26.3	18.4	24.1	17.2	21.8	16.0	19.5	24.3	18.1	22.8	16.7	20.9	2.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.8	22.1	16.2	11.6	20.5	14.7	10.5	19.2	56.1	24.4	51.6	22.9	47.2	20.9	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.3	7.4	6.6		-39.2	28.9	3.9	2.1	-41.9	30.4	-44.2	31.6	-46.4	32.7	-49.2	34.3	
				DB												
				WB	-39.2	21.2	3.8	1.5	-42.0	22.3	-44.2	23.2	-46.4	24.1	-49.2	25.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.1	-18.1	-16.9	-10.0	-3.5	3.2	10.6	14.8	11.4	6.4	-0.5	-9.7	-14.0
	DBStd	13.01	9.28	9.34	7.28	6.18	6.06	5.47	5.04	4.00	3.81	4.77	8.08	9.54
	HDD10.0	4757	872	753	620	406	228	59	12	29	119	325	591	743
	HDD18.3	7493	1130	986	878	656	469	238	134	218	358	583	841	1001
	CDD10.0	341	0	0	0	0	17	77	162	73	11	0	0	0
	CDD18.3	36	0	0	0	0	1	6	26	3	0	0	0	0
	CDH23.3	263	0	0	0	0	4	49	191	19	0	0	0	0
	CDH26.7	43	0	0	0	0	0	6	34	2	0	0	0	0
(14) Wind	WSAvg	3.4	3.3	3.3	3.6	3.5	3.7	3.6	3.3	3.2	3.3	3.6	3.3	3.3
(15) Precipitation	PrecAvg	509	34	26	24	26	37	45	62	64	59	55	41	37
	PrecMax	703	62	51	46	54	74	106	128	166	117	107	75	60
	PrecMin	324	16	3	7	7	4	11	12	12	10	16	21	8
	PrecStd	89	11	11	12	12	17	23	26	33	23	21	14	13
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	1.0	5.8	12.8	23.4	27.5	29.6	26.0	18.9	11.2	3.3	1.2
		MCWB	-0.3	0.4	2.2	6.7	13.8	18.3	20.1	19.7	14.1	9.1	2.7	0.6
	2%	DB	-2.0	-0.8	2.1	8.3	19.4	24.8	27.5	22.5	15.9	8.9	1.6	0.2
		MCWB	-2.5	-1.4	0.2	4.1	12.2	16.5	19.2	17.1	12.7	7.6	1.2	-0.2
	5%	DB	-3.8	-2.5	0.8	6.0	16.1	21.9	25.6	20.1	13.7	7.0	0.6	-0.8
		MCWB	-4.3	-3.1	-0.4	2.8	10.2	15.3	18.2	15.7	11.1	6.0	0.2	-1.3
	10%	DB	-6.1	-4.6	-0.4	4.2	13.0	19.5	23.3	18.1	11.9	5.5	-0.1	-2.3
		MCWB	-6.6	-5.2	-1.4	1.9	8.4	13.9	17.0	14.8	10.1	4.8	-0.5	-2.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	0.6	2.6	7.1	14.8	19.1	21.7	20.4	15.0	9.7	2.9	0.7
		MCDB	0.2	1.0	5.1	12.3	21.3	25.3	27.3	25.2	17.9	10.5	3.2	0.9
	2%	WB	-2.3	-1.3	0.6	4.7	12.6	17.8	20.3	18.1	13.0	7.6	1.2	-0.1
		MCDB	-2.0	-0.9	1.7	7.7	18.7	22.8	25.4	21.1	15.5	8.8	1.5	0.2
	5%	WB	-4.2	-3.1	-0.2	3.3	10.7	16.2	19.1	16.3	11.5	6.1	0.3	-1.3
		MCDB	-3.8	-2.5	0.8	5.3	15.5	20.8	23.7	19.0	13.1	6.8	0.6	-0.8
	10%	WB	-6.6	-5.2	-1.4	2.2	8.5	14.5	17.8	14.9	10.3	4.8	-0.5	-2.9
		MCDB	-6.2	-4.6	-0.4	3.9	12.9	19.1	22.6	17.7	11.7	5.5	-0.2	-2.4
(35) Mean Daily Temperature Range	MDBR		6.5	6.5	8.2	8.0	7.9	8.1	8.5	6.9	5.5	3.9	5.2	6.0
	5% DB	MCDBR	6.4	5.9	6.0	8.7	12.0	11.4	11.1	10.6	8.4	4.5	4.1	5.5
		MCWBR	6.4	5.6	4.8	5.8	6.7	6.0	5.6	6.2	5.3	3.8	4.2	5.6
	5% WB	MCDBR	6.4	5.7	5.6	7.3	11.7	10.4	10.3	9.2	7.4	4.4	4.0	5.6
		MCWBR	6.4	5.5	4.9	5.3	7.1	6.4	5.7	6.2	5.4	3.8	4.1	5.6
(40) Clear-Sky Solar Irradiance	taub		0.221	0.272	0.309	0.339	0.348	0.356	0.393	0.370	0.330	0.290	0.215	0.223
	taud		1.882	2.022	2.063	2.101	2.316	2.455	2.353	2.391	2.469	2.262	1.932	1.902
	Ebn at Noon		192	561	732	811	851	855	802	784	740	559	192	20
	Edh at Noon		20	60	95	119	107	96	104	90	66	47	17	5
(44) All-Sky Solar Radiation	RadAvg		0.09	0.67	2.15	3.84	5.01	5.03	4.93	3.19	1.72	0.58	0.15	0.02
	RadStd		0.01	0.08	0.25	0.36	0.41	0.48	0.74	0.35	0.21	0.08	0.03	0.00

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(46) Station Only		+1.28	N/A	N/A	N/A	N/A	N/A	-406	-448
(47) Regional (4 neighbors)		+1.17	N/A	N/A	N/A	N/A	N/A	-390	N/A

Nomenclature: See separate page